

Non-invasive Transcutaneous Spinal Cord Stimulation Programming Recommendations for the Treatment of Upper Extremity Impairment in Tetraplegia

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ABSTRACT

Objectives: This study analyzes the stimulation parameters implemented during two successful trials that used non-invasive transcutaneous spinal cord stimulation (tSCS) to effectively improve upper extremity function after chronic spinal cord injury (SCI). It proposes a framework to guide stimulation programming decisions for the successful translation of these techniques into the clinic.

Materials and Methods: Programming data from 60 participants who completed the Up-LIFT trial and from 17 participants who subsequently completed the LIFT Home trial were analyzed. All observations of stimulation amplitudes, frequencies, waveforms, and electrode configurations were examined. The incidence of adverse events and relatedness to stimulation parameters is reported. A comparison of parameter usage across the American Spinal Injury Association Impairment Scale (AIS) subgroups was conducted to evaluate stimulation strategies across participants with varying degrees of sensorimotor preservation.

Results: Active (cathodal) electrodes were typically placed between the C3/C4 and C6/C7 spinous processes. Most sessions featured return (anodal) electrodes positioned bilaterally over the anterior superior iliac spine, although clavicular placement was frequently used by 12 participants. Stimulation was delivered with a 10-kHz carrier frequency and typically a 30-Hz burst frequency. Biphasic waveforms were used in 83% of sessions. Average stimulation amplitudes were higher for biphasic waveforms. The AIS B subgroup required significantly higher amplitudes than did the AIS C and D subgroups. Device-related adverse events were infrequent, and not correlated with specific waveforms or amplitudes. Within the home setting, participants maintained their current amplitudes within 1% of the preset values. The suggested stimulation programming framework dictates the following hierarchical order of parameter adjustments: current amplitude, waveform type, active/return electrode positioning, and burst frequency, guided by clinical observations as required.

Conclusions: This analysis summarizes effective stimulation parameters from the trials and provides a decision-making framework for clinical implementation of tSCS for upper extremity functional restoration after SCI. The parameters are aligned with existing literature and proved safe and well tolerated by participants.

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INTRODUCTION

Upper extremity paralysis or paresis resulting from cervical spinal cord injury (SCI) severely limits independence, making restoration of hand function the highest priority for individuals with tetraplegia.^{1,2} Despite clinical practice guidelines,³ there is lack of consensus on optimal methods to restore function. Reconstructive upper extremity surgery can restore specific joint motion.^{4–6} Occupational therapy and task-oriented activity-based training aim to retrain the nervous system to recover specific motor tasks, teach compensatory strategies, and improve residual strength and range of motion.^{7–10} Assistive technologies, including orthoses or robotic exoskeletons, enable hand postures^{11,12} but do not restore voluntary movement. Noninvasive electrical stimulation techniques, such as functional electrical stimulation and neuromuscular electrical stimulation, can be applied over the skin to stimulate peripheral nerves and/or muscles directly to initiate muscle contractions in people with SCI.^{13–15} However, despite movement-based therapies, preserved neural circuits are unable to effectively drive synaptic activity in the absence of adequate descending supraspinal input, leading to a plateau in functional improvement.¹⁶

Current SCI rehabilitation approaches are constrained by the extent of functional connections between residual neural tissue and supra-spinal circuits.^{17–20} Transcutaneous spinal cord stimulation (tSCS) can target spinal cord circuits by applying currents through surface

electrodes placed on the skin over the spine.^{21–26} tSCS modulates spinal network excitability, enabling preserved descending input to the injured spinal cord to facilitate voluntary motor output.^{23,26,27} Eight studies to date have explored the use of tSCS for upper extremity rehabilitation in individuals with SCI.^{21–23,25,28–31} A recent prospective, multicenter trial (Up-LIFT—NCT04697472) revealed the ability of tSCS in conjunction with rehabilitation to improve arm and hand function after cervical SCI.³² A subsequent home-based trial (LIFT Home—NCT05284201) validated the safety and feasibility of using tSCS during take-home exercises in the home environment.

Although the Up-LIFT trial represents the largest cohort of individuals with SCI receiving tSCS in a prospective trial, best practices for programming this type of stimulation are yet to be established. This retrospective analysis reports the stimulation parameters used during the successful Up-LIFT and LIFT Home trials, assesses the stimulation parameters used in relation to previous exploratory studies, and introduces a programming flowchart to guide future clinical implementation.

MATERIALS AND METHODS

Brief Summary of the Up-LIFT and LIFT Home Clinical Trials

The Up-LIFT trial evaluated the safety and effectiveness of programmed non-invasive spinal cord stimulation (ARC^{EX} Therapy, ONWARD Medical, Inc., Boston, MA) in conjunction with

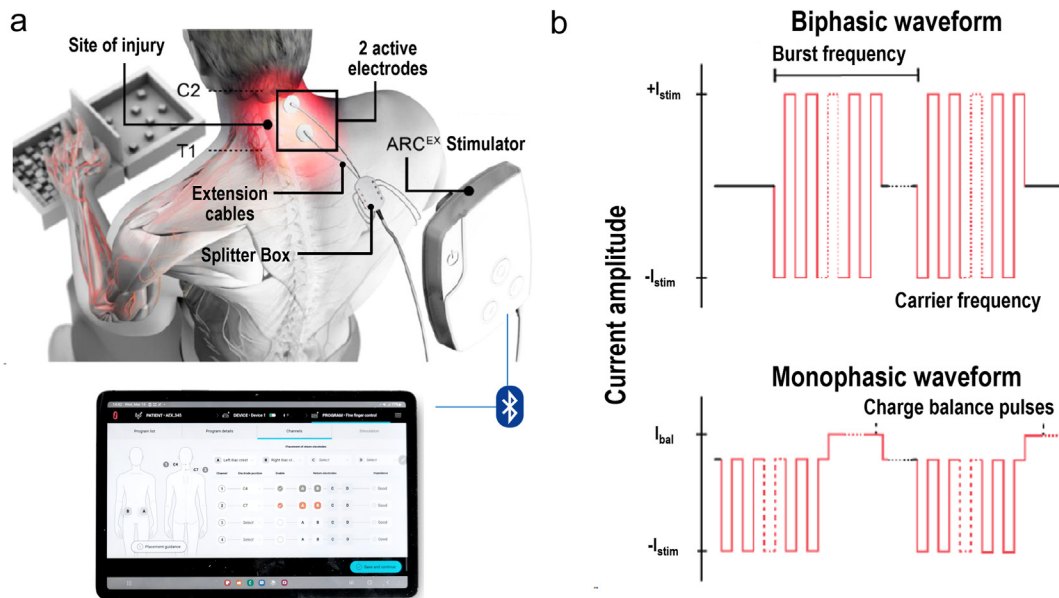


Figure 1. The ARC^{EX} System. a. The System comprises an application running on a programmer that allows the user to define or control stimulation programs delivered by the stimulator through active and return electrodes (accessories) via the splitter box and extension cables. For upper extremity training, electrodes are placed over the cervical spine. The device has a voltage-controlled output and measures impedance to ensure that the target current intensity is delivered. Programmed, noninvasive electrical spinal cord stimulation (ARC^{EX} Therapy) is delivered during in-clinic rehabilitation or during take-home exercise. b. Biphasic and monophasic waveforms both comprise bursts of rectangular pulses. The monophasic waveform includes a charge balance pulse after each burst, with charge balance amplitudes and durations dependent on the set cathodic (negative current) pulse phase amplitude. The charge balance amplitude was 12.5% of the set cathodic amplitude, producing charge balance pulse durations of 4000 μ s. [Color figure can be viewed at www.neuromodulationjournal.org]

Table 1. Up-LIFT Trial Enrollment Criteria.

Inclusion criteria (must fulfill ALL of the following):

- Aged ≥ 22 y and ≤ 75 y at the time of enrollment
- Nonprogressive cervical SCI from C2 to C8 inclusive
- ASIA AIS classification B, C, or D
- Indicated for upper extremity training procedures by participant's treating physician, occupational therapist, or physical therapist
- GRASSP-Prehension Performance score ≥ 10 or GRASSP-Strength score ≥ 30
- Minimum 12 mo postinjury
- If prescribed antispasticity or pain medications, must be at stable dose for ≥ 4 wk before commencing study procedures
- Capable of providing informed consent

Exclusion criteria (may not fulfill ANY of the following):

- Has uncontrolled cardiopulmonary disease or cardiac symptoms as determined by the investigator
- Has any unstable or significant medical condition that is likely to interfere with study procedures or likely to confound study end point evaluations such as severe neuropathic pain, depression, mood disorders, or other cognitive disorders
- Has been diagnosed with autonomic dysreflexia that is severe, unstable, and uncontrolled
- Requires ventilator support
- Has an autoimmune etiology of spinal cord dysfunction/injury
- History of additional neurologic disease such as stroke, multiple sclerosis, or traumatic brain injury
- Peripheral neuropathy (diabetic polyneuropathy, entrapment neuropathy, etc)
- Spasms that limit the ability of the participants to participate in the study training as determined by the investigator
- Received botulinum toxin injections in their upper extremity, neck, or hand within 6 mo before enrollment
- Has urinary tract infection or any of the following issues in the upper extremity or the cervical spine region at the time of enrollment: painful musculoskeletal dysfunction unrelated to SCI, unhealed fracture, contracture, pressure sore
- Breakdown in skin area that will come into contact with electrodes
- Presence of syringomyelia as confirmed by an MRI
- Currently undergoing treatment for cancer or has been in remission for < 2 y
- Received stem cell treatment within the past 2 y before enrollment
- Prior nerve or tendon transfer procedure in the upper extremities
- Total baclofen dose > 30 mg/d
- Has any active implanted medical device
- Pregnant, planning to become pregnant, or currently breastfeeding
- Concurrent participation in another drug or device trial that may interfere with this study
- Has undergone a prior course of spinal stimulation therapy directed at UE improvement
- In the opinion of the investigators, the study is not safe or appropriate for the participant, and/or the participant is unlikely to return for the follow-up visits per the protocol

ASIA, American Spinal Injury Association; MRI, magnetic resonance imaging; UE, upper extremity.

Information adapted from Moritz et al 2024³² and clinicaltrials.gov (NCT04697472).

rehabilitation, using an earlier generation of the ARCEX System (Fig. 1). The trial enrolled 65 participants with chronic, cervical incomplete SCI (American Spinal Injury Association Impairment Scale [AIS] B, C, and D) across 14 centers. A list of all inclusion/exclusion criteria can be found in Table 1. All treatment outcomes and programming guidelines presented in this study are the result of testing in individuals with SCI meeting these criteria.

A two-month "run-in phase" of upper extremity rehabilitation was followed by a two-month "treatment phase," combining tSCS and rehabilitation (tSCS+rehab). The rehabilitation program was defined to ensure homogeneity across sites. This program involved approximately one hour of training, three to four times per week, with functional task categories that included repetitive activities of gross upper extremity movement, isolated finger movements, simple and complex pinch, and grasping activities. Each participant performed one to two exercises within each category during each treatment session. Typical movement patterns were encouraged by guidance and feedback, whereas compensatory movements, such as wrist tenodesis, were strongly discouraged.

Sixty subjects completed ≥ 24 sessions (12 sessions/month) during each phase. At baseline and after each month of therapy, a battery of assessments was performed to assess upper extremity strength, function, and sensation, in addition to quality of life and

activities of daily living.³² All assessments were performed without stimulation to assess neuroplastic rather than immediate neuroprosthetic effects of tSCS+rehab. Ultimately, 72% of assessable participants were responders, achieving clinically meaningful improvements in at least one strength and one functional assessment.³² Furthermore, 90% of assessable participants achieved clinically meaningful improvements in at least one strength or functional assessment.³²

The subsequent LIFT Home trial assessed the safety of tSCS during take-home exercises in the home setting. A total of 17 participants in the previous Up-LIFT trial received a device pre-programmed with treatment parameters found to facilitate their upper extremity function during in-clinic sessions. On average, participants completed three to four at-home training sessions per week for one month. Training encompassed many of the same functional tasks as during the Up-LIFT trial, personalized to the participants' activities of daily living and using objects available in the home. More than 96% of attempted at-home training sessions were successfully completed. No serious adverse events (AEs) and no device-related AEs were reported. Importantly, effectiveness analyses revealed participants were able to further improve on the progress achieved during in-clinic rehabilitation after only one month of at-home therapy (Δ Capabilities of Upper Extremity Test

score = 1.9; Δ pinch force = 4.6 N; Δ grasp force = 6.0 N) (Tefertiller et al, in preparation). Both the Up-LIFT and LIFT Home trials received institutional review board or ethics committee approval at each site.

Initial Stimulation Parameters

Study personnel received a study manual for the LIFT device and training on device capabilities, parameter ranges, and troubleshooting. Stimulation was delivered using two round (1.25-inch diameter) active (cathodal) electrodes and two rectangular (1.5-inch $\times$ 3.5-inch) return (anodal) electrodes (Axelgaard Manufacturing Co Ltd). Stimulation comprised bursts of rectangular pulses delivered in charge-balanced monophasic or biphasic waveforms modulated by a carrier frequency (Fig. 1). The two active electrodes were supplied by two independent stimulation channels. Each stimulation channel was connected to both return electrodes through a branched cable connection. Initial therapy recommended settings were informed by early feasibility studies and existing literature (Table 2).^{21–23} These included positioning the active electrodes over the C3/C4 and C6/C7 spinous processes through palpation, placing the return electrodes bilaterally over the anterior superior iliac spine (ASIS) with the long axis parallel to the most prominent bony structure, selecting a carrier frequency of 10 kHz, a burst frequency of 30 Hz, a pulse width of 1 millisecond, and a biphasic stimulation waveform. Each site, however, had the flexibility to explore additional configurations based on individual participant anatomy, tolerance, and responses to tSCS. Before at-home use, participants and persons providing assistance also received training on device usage. To ensure effective and safe stimulation, at-home reprogramming capabilities were limited to adjusting the amplitude within $\pm 10\%$ of the value set by the researchers and informed by the participant's performance during in-clinic sessions.

Adjustment of Stimulation Parameters Throughout the Up-LIFT Trial

Baseline abilities without stimulation were assessed through quick measurements such as the time to complete a task (eg, fasten a shirt button), voluntary force using a dynamometer, and the number of task repetitions before fatigue (eg, lifting a full cup), or through evaluating movement quality (eg, handwriting). Once baseline was established, study personnel began stimulation with the recommended initial parameters above.

Therapeutic amplitudes were determined in the Up-LIFT and LIFT Home trials by gradually increasing amplitude at continuous stimulation burst frequency (typically 30 Hz) from 0 mA until the baseline ability improved or until the participant reached their tolerance for stimulation intensity. The gradual increase in intensity allowed time for some habituation to the sensation of stimulation so that individuals could tolerate intensities that may have initially been uncomfortable. Just as prior tSCS studies used posterior root muscle reflexes or twitch responses to determine stimulation amplitudes,^{21,33,34} Up-LIFT investigators could optionally determine twitch thresholds (TTs), the amplitude at which the first visible upper extremity muscle twitches occurred, and voluntary movement thresholds (VMTs), to guide therapeutic amplitude selection. Visual observation of a muscle twitch (TT) is commonly used to help identify stimulation parameters for spinal cord stimulation. VMT was defined as the amplitude at which the clinician observed or the participant reported improved performance, such as easier

Table 2. Stimulation Parameters Used in Feasibility Studies of tSCS for Upper Extremity Rehabilitation After SCI.

Publication	Burst frequency	Carrier frequency	Pulse width	Stimulation waveform	Active electrode	Return electrode	Amplitude	Duration
Inanici et al, 2021 ²¹	30 Hz	10 kHz	1 ms	Biphasic and monophasic	One above and one below level of injury	ASIS	40–90 mA	Up to 120 min/session
Inanici et al, 2018 ²²	30 Hz	10 kHz	1 ms	Biphasic	C3/C4 and C6/C7	ASIS	80–120 mA	60 $\pm$ 20 min/session
Freyvert et al, 2018 ³⁰	5–30 Hz	NA	Not specified	Not specified	C5 spinous process	ASIS	20–100 mA	60–120 min/session
Gad et al, 2018 ²³	30 Hz	10 kHz	1 ms	Biphasic and monophasic	C3/C4 and C6/C7	Iliac crests	10–250 mA	60–120 min/session
Benavides et al, 2020 ²⁹	30 Hz	5 kHz	2 ms	Biphasic	C5/C6	Iliac crests	55–95 mA	20 min/session
Wu et al, 2020 ³¹	0.2 Hz	NA	2 ms	Biphasic	T2–T4 posterior, C4/C5 anterior	Clavicles	102 mA	Not specified
Murray and Knikou, 2017 ²⁵	0.2 Hz	NA	1 ms	Not specified	Single large electrode spanning C5–T2 spinous processes	Clavicles	5–68 mA	55 $\pm$ 2 min/session
Sharma et al, 2023 ²⁸	30 Hz	5 kHz	1 ms	Biphasic	C3/C4, C5/C6, C7/T1, and T11–T12	Clavicles or iliac crests	60–90 mA	60 min/session

initiation of movement or increased readiness to move. During these optional, intermittent tests, stimulation amplitude was progressively increased with a burst frequency of 30 Hz for VMT and 0.5 to 1 Hz for TT until the threshold was obtained.

The efficacy of programming configurations was assessed over a minimum of three sessions before readjusting stimulation parameters. Study personnel evaluated functional performance within and across these sessions, often using quick assessments to determine parameter efficacy. User feedback, including participant-reported sensations, stimulation tolerance, ease of movement, muscle tone and spasticity, and secondary effects of the stimulation, such as soreness or fatigue, and new perceived changes in ability, were also considered. Safety, including device-related AEs, and particularly the maintenance of skin integrity around electrode sites, was carefully monitored. If the stimulation appeared effective after multiple therapy sessions, training continued with the selected parameters. If discomfort or no improvement was observed, one parameter was adjusted at a time to seek more effective settings, except for pulse width, which always remained at 1 millisecond. This cycle was repeated with different configurations until satisfactory settings were found.

Data Collection and Analysis

Device parameter and usage data from both the Up-LIFT and LIFT Home trials were analyzed, along with thorough reviews of participant and clinician comments and field observations. The placement of active and return electrodes, burst and carrier frequencies, and stimulation waveform usage were summarized. Amplitudes, TTs, and VMTs were summarized using descriptive statistics for all relevant observations of each waveform and each AIS subgroup. Normality of the amplitude data was confirmed using the Shapiro-Wilk test. Statistical comparisons of current amplitude across AIS subgroups were conducted using a one-way repeated-measures analysis of variance, with post hoc Bonferroni corrections for multiple comparisons. Threshold data, being nonnormal, were compared using the two-sided Wilcoxon signed-rank test. All statistical analyses were conducted using validated statistical software (Python version 3.9 with SciPy package version 1.7.3) with a significance level of 0.05. Data are reported as mean $\pm$ standard deviation.

RESULTS

Active Electrode Placement

Active electrode placement with one electrode placed rostrally and another electrode caudally to the injury level were shown to be effective in feasibility studies.^{21–23} Given the high incidence of injuries near the C4 to C5 level,³⁵ it was initially advised to position the upper and lower active electrodes on the midline above the C3/C4 and C6/C7 spinous processes, respectively. Adherence to these guidelines was high, with 52% of all observations using the C3/C4 and C6/C7 active electrode placements, irrespective of return placement. Although other active electrode locations were tested, they remained within the recommended levels of C3 to T1 (rostral range: C3/C4–C6/C7; caudal range: C5/C6–C7/T1). Distribution of rostral and caudal active electrode placements was similar whether return electrodes were placed over the ASIS or clavicles (Fig. 2).

Return Electrode Placement

Although most sessions (89%) used bilateral return electrodes over the ASIS, several sessions explored alternate placement over the clavicles (Table 3). Similar to placement over the ASIS, the long axes of

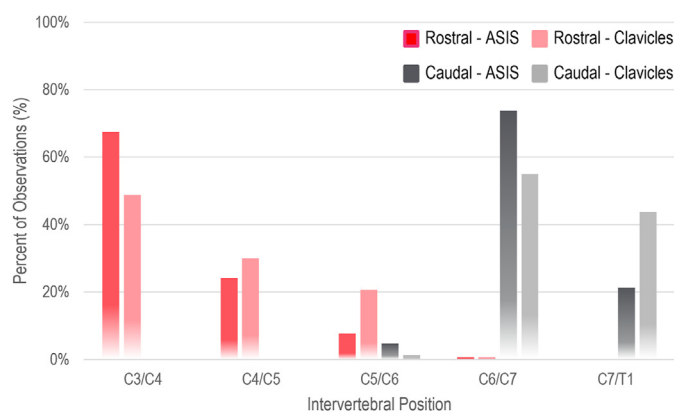


Figure 2. Active electrode placements across Up-LIFT sessions. A total of 1502 instances of reported active electrode configurations were used, with 1342 instances using ASIS return electrode placement and 160 instances using clavicular return electrode placement. Placement of rostral and caudal active electrodes with return electrode placed over the ASIS (darker pink and gray bars) and clavicles (lighter pink and gray bars). [Color figure can be viewed at www.neuromodulationjournal.org]

the rectangular return electrodes were placed parallel to the orientation of the clavicle bones in these instances. Clavicle placement was primarily attempted in nonresponsive participants who continued to exhibit slow progress after other stimulation program modifications or who experienced aversion to stimulation with return electrodes over the ASIS. Of 60 participants, 18 (30%) used clavicle placement in at least one session, with 11 participants (61%) expressing a preference for clavicle placement over ASIS owing to improved sensation/stimulation tolerance ($n = 6$, 55%) or improved function ($n = 5$, 45%). Only one participant attempting clavicle placement reported reduced comfort. For the remaining participants, clavicle placement was tolerable but did not lead to substantial changes in performance, so the original ASIS placement was resumed.

The utilization of bilateral clavicle placement varied depending on preference and the timing of its introduction, with 12 participants using it in $>20\%$ of their total sessions. The proportions of participants in AIS B, C, and D subgroups who frequently used clavicle return electrode placements were 56% (5/9), 21% (6/28), and 4% (1/23) respectively. Among frequent users of clavicle placement, seven participants (58%, all AIS C or D) were classified as responders at study completion.

Stimulation Parameters

Burst and Carrier Frequencies

The stimulator enabled adjustment of burst and carrier frequencies. On the basis of feasibility studies (Table 2), a burst frequency of 30 Hz and carrier frequency of 10 kHz were initially recommended. All participants used the 10-kHz carrier frequency for all exercises, and $>98\%$ of exercise observations used the 30-Hz burst frequency (Table 4).

Although several participants attempted exercises with varying burst frequencies (from 20–100 Hz), only two continued using these different burst frequencies during $\geq 10\%$ of their recorded exercises. These participants reported excessive tone and spasticity, which hindered active participation in therapy sessions, prompting investigators to explore alternate frequencies. Frequencies >30 Hz were noted to decrease tone and facilitate participation in rehabilitation.

Table 3. Use of Clavicles for Return Electrode Placement During the Up-LIFT Trial.

Use of clavicles for return electrode placement	<i>n</i>	%
Total tSCS+rehab training sessions	1504	
Sessions with return electrodes on clavicles	162	11%
Total participants (modified intent-to-treat population*)	60	
Participants testing return electrodes on clavicles	18	30%
Participants indicating clavicles preferred	11	61%
Preference due to:		
Improved sensation	6	55%
Improved function	5	45%
Participants documenting discomfort with clavicles	1	5.6%
Participants with frequent clavicle use (>20% sessions)	12	67%
Frequent clavicle users who were responders	7	58%
AIS C	6	86%
AIS D	1	14%
Frequent clavicle users who were nonresponders	5	42%
AIS B	5	100%

*Modified intent-to-treat population includes all individuals who completed ≥ 24 sessions (12/mo) of rehab alone, and ≥ 24 sessions (12/mo) of tSCS+rehab.

Stimulation Waveforms

Both biphasic and monophasic waveforms were charge-balanced, with monophasic waveforms including a comparatively long, low amplitude return phase after each burst (Fig. 1). Biphasic waveforms provide milder stimulation with a more gradual ramp-up in stimulation amplitude, which may limit the development of excessive muscle tone that may interfere with motor function. Therefore, Up-LIFT investigators were advised to begin with biphasic stimulation but had the flexibility to test either waveform. Most exercises (83%) during the Up-LIFT trial used biphasic stimulation (Fig. 3).

Current Amplitude

Therapeutic current amplitudes were identified on the basis of visual observations of motor performance and participant feedback, with frequent adjustments within and between sessions. The average amplitudes used across all exercises and participants were 39.3 ± 25.5 mA and 52.4 ± 31.0 mA for monophasic and biphasic stimulation waveforms, respectively (Fig. 3, $p < 0.001$).

Notably, the AIS B subgroup used higher therapeutic amplitudes than did the AIS C and D subgroups for both monophasic (Δ mean = 22.7 ± 40.0] and 22.0 ± 41.3 , respectively) and biphasic stimulation (Δ mean = 16.1 ± 45.0 and 26.7 ± 42.3 , respectively) (all $p < 0.001$). AIS C and D subgroups differed significantly in

amplitude only within the biphasic waveform (Δ mean = 10.6 ± 41.3] mA; $p < 0.001$) (Fig. 4).

Twitch Thresholds and Voluntary Movement Thresholds

The optional TTs and VMTs were recorded for at least one session in 33 of 60 participants (55%) across eight of the 14 sites (57%). Average TT and VMT values were 40.6 ± 12.8 mA and 50.8 ± 17.9 mA, respectively, for monophasic stimulation, and 64.3 ± 33.3 mA and 59.6 ± 35.5 for biphasic stimulation (Fig. 5). The recorded mean VMT values were significantly higher than the TT values for monophasic stimulation but significantly lower for biphasic stimulation (all $p < 0.001$).

Parameters in Use During Device-Related AEs

The incidence of device-related AEs occurring during stimulation was low, and these events were only mild to moderate in severity and promptly resolved when stimulation was turned off or current amplitude was reduced. To determine whether specific stimulation parameters were associated with a higher likelihood of AEs, we examined the waveform and average current amplitudes within sessions in which “at least possibly” device-related AEs occurred (Table 5). There were no serious AEs related to the device or study procedures in either trial. In addition, the incidence of nonserious, device-related AEs was low (1.5% of all tSCS+rehab sessions). The distribution of monophasic and biphasic waveforms used during these occurrences mirrored the overall waveform usage in the study. Stimulation amplitudes used during these occurrences were lower than the average amplitudes used for each waveform, and significantly below the maximum amplitudes tested throughout the study.

Programming Adjustments in the Home Setting

In the LIFT Home trial, stimulators were preconfigured with up to two home-use therapy programs based on parameters found to be comfortable and effective during in-clinic sessions. Although participants primarily used program 1, program 2 allowed an alternate waveform or amplitude range for specific exercises as needed. The training dose completed by participants in LIFT Home was equivalent to one month of in-clinic therapy (Table 6).

Table 4. Summary of Burst and Carrier Frequencies Used in the Up-LIFT Trial.

Burst Frequency (Hz)	Observations (%)	Participants (%)
20	4 (0.04%)	2 (3.3%)
30	8893 (98.83%)	60 (100%)
60	47 (0.52%)	3 (5%)
90	40 (0.45%)	3 (5%)
100 Hz	3 (0.03%)	1 (1.7%)
All others (30–90 Hz)	11 (0.12%)	8 (13%)
Carrier frequency (kHz)	Observations (%)	Participants (%)
10	8998 (100%)	60 (100%)

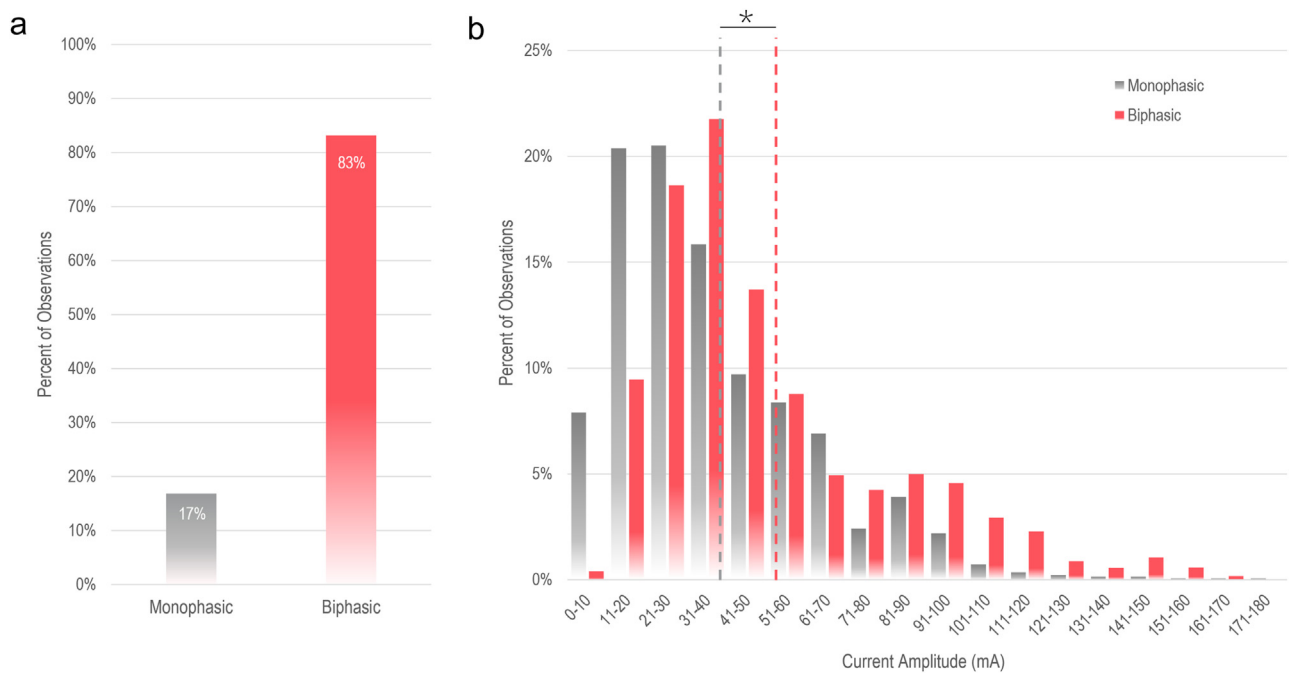


Figure 3. Waveform usage. a. Monophasic and biphasic stimulation waveform usage within the Up-LIFT trial (number of observations = 8998). b. Distribution of amplitudes used with monophasic and biphasic waveforms. All reported amplitudes represent the cathodic pulse phase amplitude for each waveform. Vertical dashed lines represent the mean amplitudes for each waveform, calculated as the average amplitudes across both active electrodes for all participants and all relevant exercise observations within each waveform (monophasic: number of observations = 3042, number of participants = 36; biphasic: number of observations = 14,954, number of participants = 58). Mean monophasic and biphasic amplitudes were significantly different ($p < 0.001$). [Color figure can be viewed at www.neuromodulationjournal.org]

Participants typically did not need to adjust the preset current amplitudes. Considering all sessions, the amplitudes delivered at home through both programs remained within $1\% \pm 4\%$ of the investigator-determined, prespecified values. When adjustments were made, increases in amplitude within the allowed 10% range

were more common than decreases. Average increases were $9\% \pm 2\%$, and average decreases were $6\% \pm 3\%$ across both programs. No participant required set-point reprogramming, indicating the preprogrammed settings were well tolerated at home and that participants were generally satisfied with the values identified by

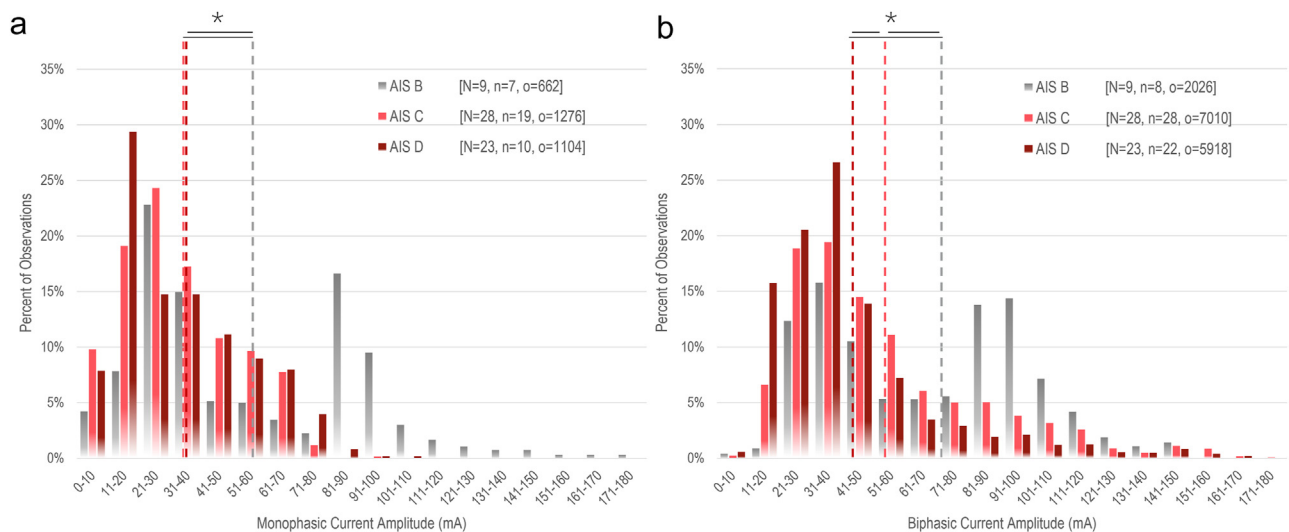


Figure 4. Amplitude usage by AIS subgroup. Panel a shows monophasic and panel b biphasic waveforms. Vertical dashed lines represent the mean amplitudes for each subgroup, calculated as the average amplitudes across both active electrodes for all participants and all relevant exercise observations within each waveform. N refers to the total number of participants within each AIS subgroup, n refers to the number of participants contributing to the observations within each subgroup and waveform, and o refers to the number of observations. *Statistically significant difference among designated AIS subgroups (all $p < 0.001$). [Color figure can be viewed at www.neuromodulationjournal.org]

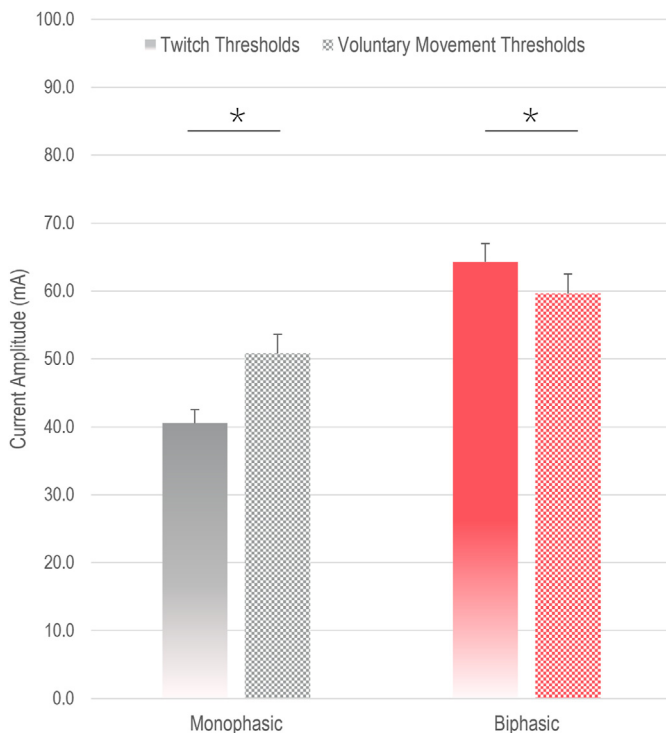


Figure 5. Twitch thresholds (TTs) and voluntary movement thresholds (VMTs) identified in the Up-LIFT trial. Thresholds are averaged across all relevant measurements within each waveform (TT monophasic: number of observations = 40, number of participants = 9; TT biphasic: number of observations = 156, number of participants = 33; VMT monophasic: number of observations = 41, number of participants = 10; VMT biphasic: number of observations = 152, number of participants = 31). Error bars indicate standard error of the mean. * $p < 0.001$. [Color figure can be viewed at www.neuromodulationjournal.org]

study personnel through their sessions in the Up-LIFT trial. Still, 65% of LIFT Home participants (11/17) made at least one adjustment at home, indicating ease of use and confidence with self-adjustment. Only two incidences of extra research clinic visits were required to adjust stimulation parameters, and these were specific to the stimulation waveform within a single at-home therapy program.

DISCUSSION

The Up-LIFT and LIFT Home trials showed the safety and effectiveness of tSCS combined with rehabilitation or take-home exercises in improving upper extremity strength and function in individuals with chronic incomplete cervical SCI. No serious device-related AEs were reported during the trials, and nonserious device-related AEs did not correlate with any particular stimulation parameter used.

Active Electrodes

Multisite stimulation was favored in the Up-LIFT and LIFT Home trials, specifically positioning active electrodes between C3/C4 and C6/C7 spinous processes. This electrode montage spans the cervical enlargement, allowing current delivery to many neural circuits involved in upper extremity function. Multisite stimulation is also effective for distal muscle contraction and enhances motor evoked potential responses.²⁴ Preliminary studies have similarly shown improved motor function with multisite stimulation.^{21–23} Although

recent research shows electrode arrays aligned with the segmental location of cervical motor pools increase the relative amplitude of specific evoked responses,³⁶ the clinical utility of this feature has not been methodically addressed for people with SCI. Moreover, anatomical variations pose challenges in determining precisely which segmental neural circuits are being stimulated, with lack of specificity an inherent property of tSCS.³⁷ Indeed, this therapy is not meant to drive specific muscle group functions but to enhance basal excitability to enable voluntary activation of preserved neuronal substrates, and lack of specificity does not preclude the clinical value of the intervention. Therefore, active electrode repositioning is a lower priority for adjustments and reserved mainly for instances when discomfort or poor electrode adherence due to prior scars are evident.

Return Electrodes

Although most prior tSCS studies for upper extremity rehabilitation in SCI used ASIS placement for return electrodes, some studies have indicated successful evoked upper extremity responses with return electrodes positioned over the clavicles (Table 2).^{25,28,31} It was hypothesized that this clavicle configuration may provide a more focused current path that could potentially benefit participants who did not sufficiently respond otherwise.

Return electrode placement over the clavicles was well tolerated and seemed beneficial for participants unresponsive to tSCS when the electrodes were placed over the ASIS. Although the manipulation of return electrode placement to the clavicles allowed all participants who were previously unresponsive in the AIS C or D groups to meet the predefined responder criteria, this was not the case for individuals with more severe impairments in the AIS B group. It is important to note that the AIS B subgroup was the least responsive to tSCS+rehab overall.³² This suggests that the lack of response is more likely related to the severity of their injuries and significantly reduced preservation of residual descending neural connectivity when compared with the other subgroups. It is likely that individuals with more severe injuries may require more frequent and significant adjustments in stimulation parameters to achieve observable improvements. These findings should be interpreted cautiously owing to small sample sizes and confounding factors. The recommendation for initial return electrode placement remains over the ASIS on the basis of success in feasibility studies and for most participants in the Up-LIFT and LIFT Home trials, but placement on the clavicles may be explored as needed should other parameter adjustments not suffice.

Stimulation Frequencies

The 10-kHz carrier frequency was consistently used in all stimulation sessions during the Up-LIFT and LIFT Home trials.^{27,38} Although the use of alternative carrier frequencies was not systematically investigated, the low incidence of device-related AEs, particularly those related to pain or discomfort, across both trials supports the safety and recommendation for continued use of this parameter.

Although most stimulation sessions used a 30-Hz burst frequency, higher burst frequencies were tested for participants experiencing hand spasticity and increased tone hindering therapy participation. Higher frequencies have been reported to decrease spasticity in individuals with SCI.^{39,40} Higher burst frequencies (50 Hz) can decrease exaggerated reflex responsiveness,³⁹ decrease clonus and cutaneous-input-evoked spasms evoked by cutaneous

Table 5. Stimulation Parameters Associated With Device-Related AEs in the Up-LIFT Trial.		
	Count (% of sessions)	Participants (% of mITT population)
Total tSCS+rehab sessions	1504 (100%)	60 (100%)
tSCS+rehab sessions with AE possibly device-related	23 (1.5%)	11 (18%)
	Count (% of device-related AEs)	Count (% of patients experiencing AEs)
Causality:		
Possibly related*	9 (39%)	6 (55%)
Probably related†	12 (52%)	5 (45%)
Causal relationship‡	2 (9%)	2 (18%)
AE MedDRA preferred term:		
Autonomic dysreflexia†	2 (9%)	1 (9%)
Bladder spasm*†	4 (7%)	1 (9%)
Burning sensation†	1 (4%)	1 (9%)
Dizziness*	2 (9%)	1 (9%)
Erythema‡	1 (4%)	1 (9%)
Headache†	2 (9%)	2 (18%)
Hyperhidrosis*†	3 (13%)	3 (27%)
Malaise†	1 (4%)	1 (9%)
Medical device site reaction*	1 (4%)	1 (9%)
Nausea†	1 (4%)	1 (9%)
Pain*‡	2 (9%)	2 (18%)
Respiratory rate increased†	1 (4%)	1 (9%)
Skin irritation†	1 (4%)	1 (9%)
Urinary incontinence*	1 (4%)	1 (9%)
Stimulation waveform during device-related AEs		
Monophasic	5 (22%)	4 (36%)
Biphasic	18 (78%)	7 (64%)
Stimulation amplitudes during device-related AEs	Monophasic	Biphasic
Mean (SD)	22.7 (13.5)	36.6 (9.7)
Median	24.2	33.5
Min, max	5.8, 37.9	17.9, 51.7
mITT population includes all individuals who completed ≥24 sessions of rehab alone, followed by ≥24 sessions of tSCS+rehab.		
MedDRA, Medical Dictionary for Regulatory Activities; mITT, modified intent-to-treat.		
*Possibly related; †probably related; and ‡causal relationship to device use.		

input,⁴¹ and decrease hyperreflexic stretch responses⁴² in people with SCI. Although the use of higher frequencies was limited to two participants with problematic spasticity, the observed results and the potential facilitation of motor function through decreasing muscle tone are supported by existing literature.

Relationships Between Waveform, Amplitude, and Thresholds

Transcutaneous biphasic stimulation may be superior in facilitating fine motor control tasks, whereas monophasic stimulation

may be more effective at improving strength.²¹ Interestingly, the recorded VMTs were significantly higher than the TTs for monophasic stimulation, whereas they were significantly lower for biphasic stimulation (Fig. 5). This suggests that biphasic stimulation may facilitate voluntary movement before inducing unwanted twitch contractions, supporting the initial use of the biphasic setting for therapy.

Appropriate determination of stimulation amplitudes relative to motor thresholds is debated. Submotor, motor, and

Table 6. Duration of tSCS+rehab During the Up-LIFT and LIFT Home Trials.				
	Sessions/mo	Total sessions	Duration of stimulation/session (HH:mm)	Cumulative stimulation duration (HH:mm)
Up-LIFT (2-mo tSCS+rehab phase)				
Mean (SD)	12.5 (1.2)	25.1 (1.5)	1:00 (0:05)	25:10 (1:54)
Median	12	24	1:00	24:14
Min, max	9, 16	24, 29	0:25, 1:29	21:30, 30:31
LIFT Home (1-mo tSCS+rehab phase)				
Mean (SD)	12.3 (2.9)	12.3 (2.9)	0:59 (0:10)	12:08 (3:10)
Median	12	12	1:00	12:08
Min, max	6, 20	6, 20	0:41, 1:13	4:28, 18:15

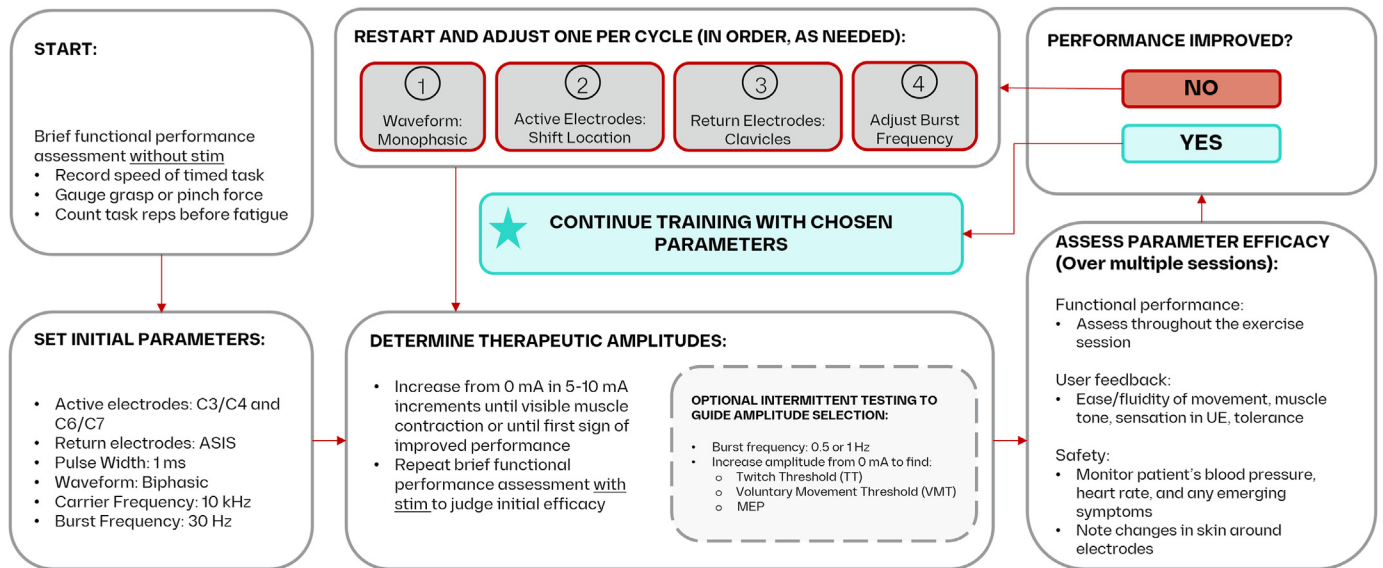


Figure 6. tSCS parameter determination flowchart based on strategies used in the successful Up-LIFT trial. MEP, motor evoked potential; UE, upper extremity. [Color figure can be viewed at www.neuromodulationjournal.org]

supramotor thresholds have all been reported as methods to optimize particular upper extremity responses.⁴³ One study showed that using submotor threshold stimulation led to improved pinch force, The Graded Redefined Assessment of Strength, Sensibility and Prehension (GRASSP) sensibility, and GRASSP prehension abilities for individuals with SCI.²¹ A separate study found using amplitudes equivalent to motor threshold led to improvements in grip strength.²³ Still others have indicated positive outcomes in upper extremity functions when using supramotor threshold stimuli.^{31,34} In both the Up-LIFT and LIFT Home trials, stimulation amplitude was not restricted to pre-determined motor thresholds. Instead, it was individualized on the basis of visual observations and feedback. Amplitudes used in the AIS B subgroup were significantly higher than in participants in AIS C and D, indicating a potential need for more intense stimulation in individuals with more severe SCI. This aligns with previous studies, in which participants in AIS B generally required higher current amplitudes.²¹ This trend also suggests that participants in AIS B might tolerate higher amplitudes before experiencing discomfort, possibly owing to differences in sensory sparing. Importantly, stimulation amplitude did not correlate with the incidence of potentially device-related AEs, reinforcing the safety and tolerance across the full range of the stimulation amplitudes used in the trials (1–180 mA).

Comparisons of stimulation parameters (amplitudes, waveforms, active and return electrode placements) used by responders and nonresponders in the Up-LIFT trial were performed, with no significant differences found. Because 99% of observations used a 30-Hz burst frequency and all observations used a 10-kHz carrier frequency, a comparison of frequencies in responders and nonresponders was not feasible. A lack of clear distinction between universally effective and noneffective parameters is unsurprising owing to the heterogeneity of the participant population, differences in sensory preservation, skin impedance, and anatomy, which can affect effective stimulation parameters.

Programming Recommendations

The initial programming guidance and unique site-by-site processes used to determine effective stimulation parameters in each participant were informed by personnel with expertise in tSCS and SCI rehabilitation. Combining this historical knowledge with the resultant successful programming outputs previously outlined, an updated tSCS programming flowchart has been created to guide future implementation of this technology (Fig. 6). Amplitude remains the most frequently manipulated parameter. Should simple amplitude adjustments fail to facilitate meaningful improvement, we suggest reducing the amplitude and changing the waveform to monophasic, given clinical effectiveness was shown within the clinical trials with both waveforms. We then suggest adjusting in order as needed: active electrode location, return electrode location, and finally, burst frequency. Burst frequency is the final suggested adjustment because to date, there is a lack of existing evidence supporting upper extremity improvement as a function of burst frequency, whereas adjustments in other parameters were shown to be beneficial in the trials.

CONCLUSIONS

The Up-LIFT and LIFT Home trials represent a significant milestone in the field of neuromodulation by indicating the safety, usability, and effectiveness of tSCS therapy for recovery after SCI in clinical and home environments. SCI is an intricate condition, with various injury characteristics determining the trajectory of neurologic recovery, extent of SCI secondary health conditions (ie, spasticity), and overall treatment plans (pharmacologic, surgical, etc). Trial exclusion criteria aimed to minimize this variability; however, the need to individually optimize stimulation strategies across various specific deficits remains apparent. The proposed workflow provides structured guidance, incorporating practices that showed beneficial outcomes within the heterogeneous study

population. The intention is not to prescribe a rigid protocol but rather to assist users with a refined decision-making algorithm that will facilitate the implementation of this novel intervention in the clinic and home settings.

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Authorship Statements

Kristen Gelenitis and Jessica M. D'Amico performed the study analysis and interpretation of results. Kristen Gelenitis, Andrea Santamaria, and Jessica M. D'Amico prepared the manuscript. Jared Pradarelli and Markus Rieger assisted in revising the flow diagram to its final form on the basis of their experience as ONWARD field clinical specialists involved in the Up-LIFT and LIFT Home trials. Fatma Inanici created the initial version of the programming flow diagram that helped guide investigators during the Up-LIFT trial. Edelle Field-Fote, James Guest, Chet Moritz, and Candace Tefertiller were investigators during the Up-LIFT trial and collaborated on manuscript editing and refinement. Jenny Suggitt and Amanda Turner were study personnel for the Up-LIFT and LIFT Home trials. All authors approved the final manuscript.

Conflict of Interest

Kristen Gelenitis, Jared Pradarelli, Jenny Suggitt, Amanda Turner, and Markus Rieger are employees of ONWARD Medical. Andrea Santamaria, Jessica M D'Amico, Edelle Field-Fote, and Chet Moritz are paid consultants of ONWARD Medical. The remaining authors reported no conflict of interest.

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COMMENT

This research provides analysis of, in addition to promising details and insights for guidance in, the use of tSCS therapy in a patient population when there are limited treatment options for upper extremity functional recovery in patients with SCI.

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